MOSFET Fundamentals – Week 2

Fire EV Internship | CMOS & VLSI

During this week, I learned about MOSFETs and the working of NMOS and PMOS transistors. I studied their three operating regions: cutoff, linear and saturation. I also learned how gate-source voltage and drain-source voltage affect drain current, along with the effect of the W/L ratio and channel-length modulation.

For practical work, I used ngspice with the SKY130A PDK to simulate NMOS and PMOS transistors. I generated their output and transfer characteristic graphs using Python. The output graphs show how drain current changes with drain-source voltage for different gate voltages. The transfer graphs show how drain current increases as the gate voltage rises beyond the threshold region.

I also estimated the threshold voltage using the constant-current method. The estimated values were 0.6169 V for NMOS and –0.5897 V for PMOS. Overall, this week helped me connect the concepts I studied with practical transistor simulations and understand how to read their characteristic curves.

Checkpoint Questions and Answers

Q1. What physically happens at the threshold voltage?

When the gate voltage reaches the threshold voltage, enough electrons gather beneath the gate oxide to form an inversion channel between the source and drain. This allows significant current to flow when a drain-source voltage is applied.

Q2. How does doubling W change the drain current, and why?

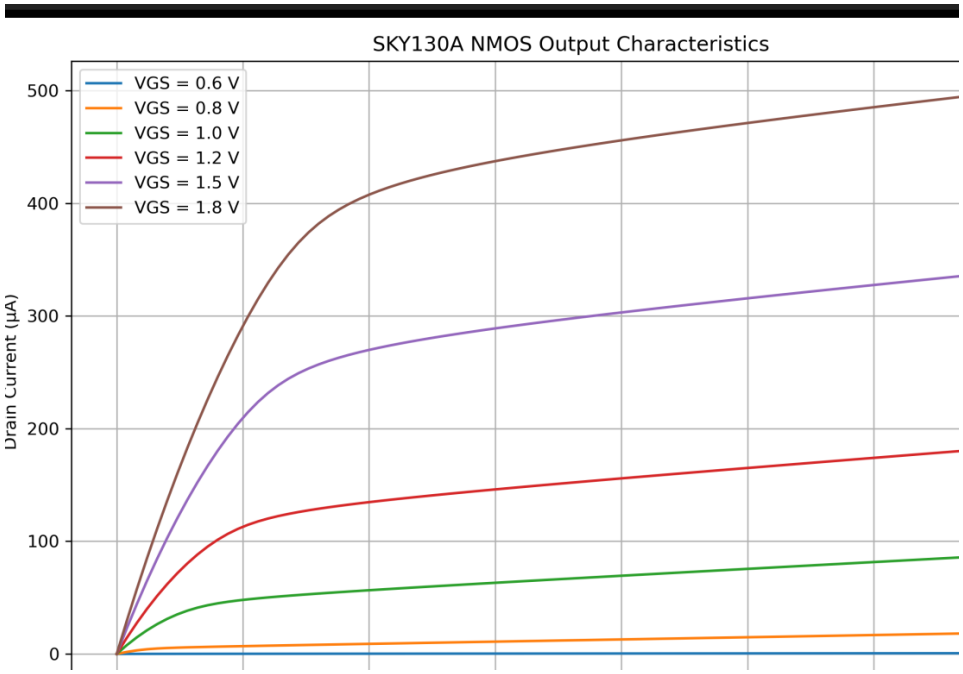
If the channel length and operating conditions remain the same, doubling the width approximately doubles the drain current in the ideal MOSFET model. A wider channel allows more charge carriers to flow.

Q3. Why is a PMOS usually made wider than an NMOS?

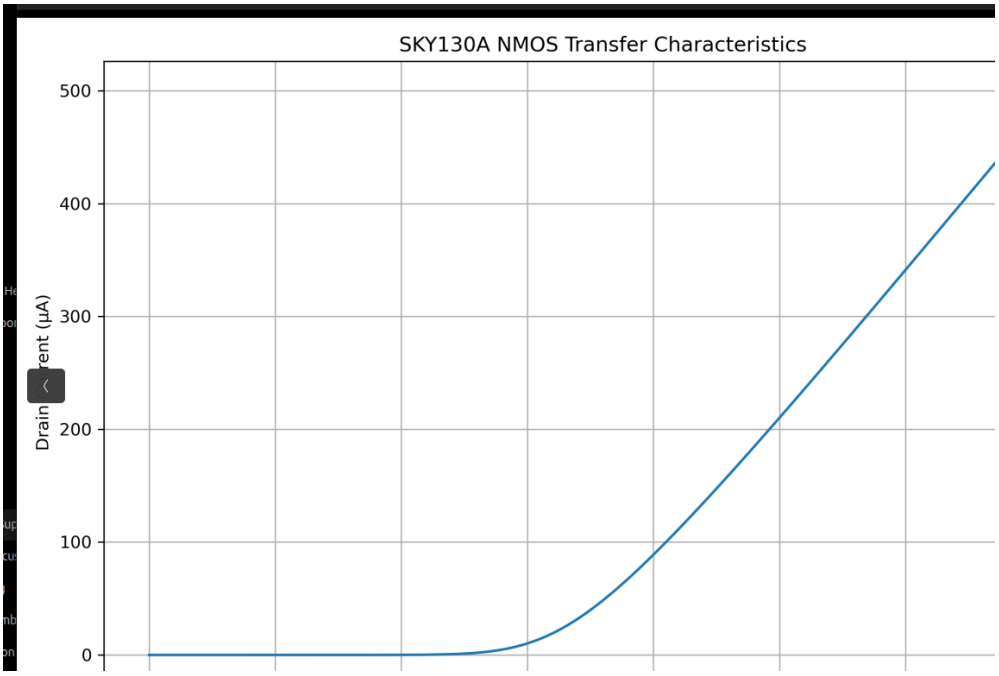
The mobility of holes in a PMOS is lower than the mobility of electrons in an NMOS. A wider PMOS helps compensate for this difference and provides comparable drive strength.

Simulation Plots

The following graphs were generated from the NMOS and PMOS simulations using ngspice and the SKY130A models.

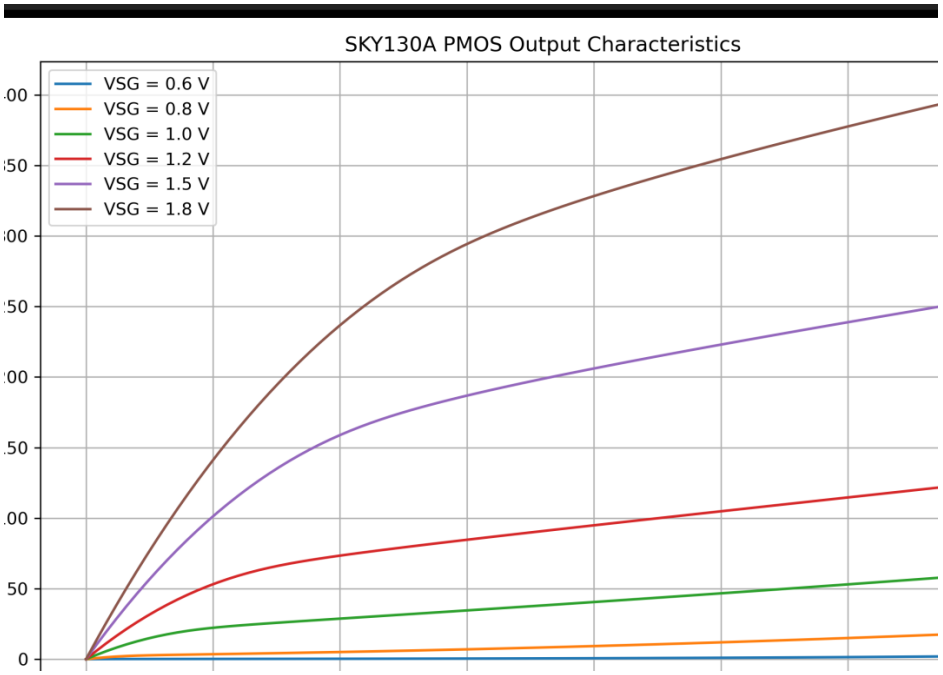


NMOS Output Characteristics – Drain current (ID) versus drain-source voltage (VDS) for different gate-source voltages (VGS).

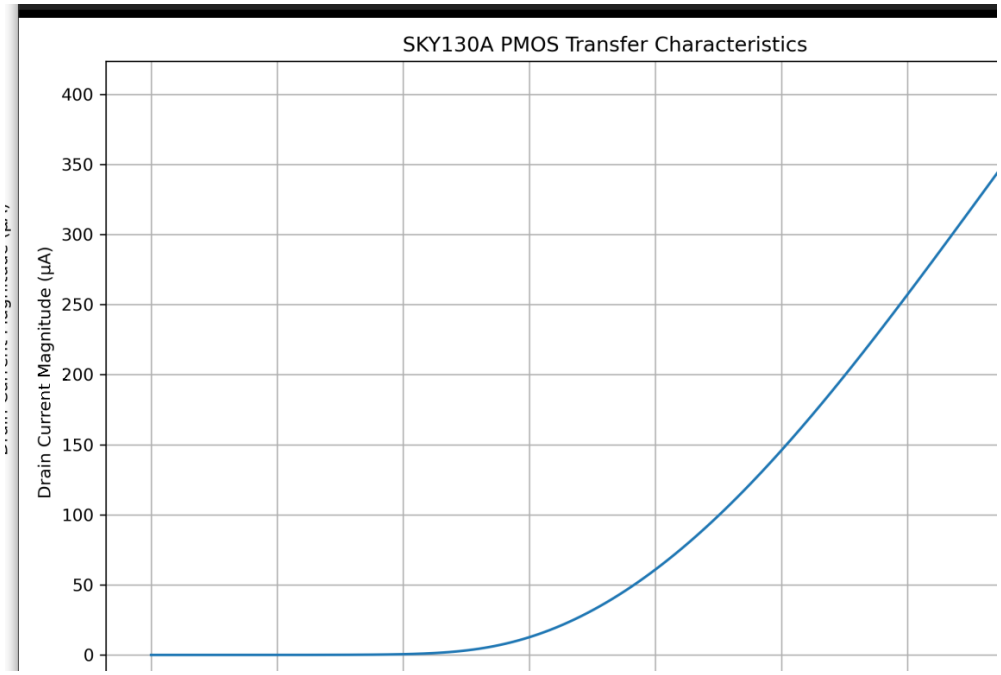


NMOS Transfer Characteristics – Drain current (ID) versus gate-source voltage (VGS).

Simulation Plots (continued)



PMOS Output Characteristics – Drain-current magnitude versus source-drain voltage magnitude (VSD) for different VSG values.



PMOS Transfer Characteristics – Drain-current magnitude versus source-gate voltage magnitude (VSG).